

Question	Working space	Mark	Comments
1	${}^0_{-1}\text{e}$ ✓ ${}^{42}_{20}\text{Ca}$	2	Accept e^- , β^- or β with the correct numbers. Not b or B
Total		2	

Question	Marking guidance	Mark	Comments
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Question	Marking guidance	Mark	Comments
07.1	2.5 y to 2.8 y ✓ Construction seen on graph ✓ Repeat and average seen ✓	3	
07.2	Paper moves between source and detector ✓ Count rate OR reduction of count rate OR absorption measured ✓ If count rate too high, rollers separate OR if count rate too low, rollers move together ✓ Either beta suitable because will pass through paper with some absorption OR half-life suitable because activity will not change quickly OR long half-life so does not need frequent replacement ✓	4	
Total		7	

Question	Marking guidance	Mark	Comments

Question	Marking guidance	Mark	Comments
10.1	Well-drawn line, labelled s, indicating the distance <u>between the top ball and the bottom ball</u> ✓	1	s should be from the top of the 1 st ball to the top of the last ball or from the bottom of the 1 st ball to the bottom of the last ball
10.2	largest distance gives smallest (percentage) uncertainty ✓	1	Accept smaller (percentage) error
10.3	198 ± 1 (mm) ✓ ecf	1	Accept answer in m or cm. Must have value and unit correct and consistent If the candidate has marked s between a different pair of balls they can still have the mark. The other acceptable distances are: 1 st to 2 nd = 8 (mm) 1 st to 3 rd = 32 (mm) 1 st to 4 th = 71 (mm) 1 st to 5 th = 127 (mm) 2 nd to 3 rd = 24 (mm) 2 nd to 4 th = 63 (mm) 2 nd to 5 th = 118 (mm) 2 nd to 6 th = 190 (mm) 3 rd to 4 th = 39 (mm) 3 rd to 5 th = 94 (mm) 3 rd to 6 th = 162 (mm) 4 th to 5 th = 56 (mm) 4 th to 6 th = 121 (mm) 5 th to 6 th = 71 (mm) The same uncertainties and requirement for correct unit apply to the ecf

Question	Marking guidance	Mark	Comments
10.4	Use of $s = \frac{1}{2}gt^2$ with candidate's value for s leading to 9.6 to 10.1 (m s ⁻²) to 2 or 3 sf ✓	1	Candidates who do not measure s from the 1 st ball will have to do a lot of maths to access the 2 nd mark Look for use of $s = ut + \frac{1}{2}gt^2$
10.5	$\frac{1}{198} \times 100$ leading to 0.5 % ✓	1	Accept appropriate value for candidate's 10.1 and 10.3 Ignore sf
10.6	Adds some percentage uncertainties ✓ $2 \times 0.8(\%) + \text{candidate's } 10.5 \text{ ✓}$ (±) 0.2 or 0.21 ecf ✓	3	Can access the 3 rd mark for correctly calculating the absolute uncertainty from their own (incorrect) percentage uncertainty
Total		8	

Question	Marking guidance	Mark	Comments
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question	marking guidance	mark	comments
05.1	Any mention of neutrino (1 mark) ✓ Electron-neutrino OR ν_e (2 marks) ✓	2	$\bar{\nu}_e$ OR anti-electron neutrino scores max 1
05.2	Two half-lives so the F-18 has quartered ✓ $\frac{25\%}{75\%} = 0.33$ ✓	2	Accept 1:3, 33%, $\frac{1}{3}$ Do not credit $\frac{1}{4}$ without working seen
05.3	The activity (of the sample) must be large enough ✓ to be distinguishable from background radiation OR so that it is not all absorbed by the body (OWTTE) ✓	2	Condone idea that too much of the F-18 will have decayed (in a short time) for MP1
Total		6	

Question	Marking guidance	Mark	Comments
	Equal in magnitude	1	Accept:

07.1	$5.5 \times 10^6 \times 1.6 \times 10^{-19}$ seen OR 8.8×10^{-13} (J) seen ✓ Use of $E_k = \frac{1}{2}mv^2$ ✓ Leading to $1.63 \times 10^7 \text{ m s}^{-1}$ ✓	3	“Use of” means correct rearrangement or substitution Value given to at least 3 sf
07.2	Substitution into $m_x v_x = (-)m_\alpha v_\alpha$ $(3.6 \times 10^{-25} \times v_x = 6.6 \times 10^{-27} \times 1.63 \times 10^7)$ OR $218v_x = 4v_\alpha$ ✓ $v_x = 3.0 \times 10^5 \text{ m s}^{-1}$ ✓	2	Condone POT error in MP1 $1.6 \times 10^7 \text{ m s}^{-1}$ gives $2.9 \times 10^5 (\text{m s}^{-1})$
07.3	Conservation of momentum or energy discussed ✓ Third particle carries some energy or momentum ✓ Each decay (of a given nuclide) has the same energy ✓ Mention of components of momentum (perpendicular to direction of recoil) ✓	Max 3	Accept converse
Total		8	

Question	Marking guidance	Mark	Comments
	(Horizontal $u = 48 \cos 15^\circ \text{ ssn} (= 46.4 \text{ m s}^{-1})$ ✓		

11.1	reasonable precaution ✓ comments on precaution in terms of irradiation or contamination ✓	–	Use long tongs ✓ To reduce irradiation to the body ✓ OR Wear gloves or wash hands after use ✓ To reduce chance of contamination ✓ OR Minimise the time the source is out of its container ✓ To reduce irradiation to the body ✓
11.2	23 ± 2 counts per minute ✓	1	Expect to see a smooth curve drawn on the graph
11.3	Construction of best-fit line ✓ Deduct background OR Halve assigned corrected count rate ✓ Answer within range (1.3 ± 0.3) minutes ✓	3	Ecf from 11.2 Time is 78 ± 18 s Award max 2 for candidate who treats the data as already corrected for background, ie ≈ 1.9 s

11.4	Named suitable counting apparatus✓ Operated with the source not present✓ Method to ensure accuracy of measurement✓	3	Eg Geiger(-Müller) counter/tube/ ratemeter scaler counter E.g reading for long time repeat and average readings
Total		9	

Question	Key
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25	D
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Question	Marking guidance	Mark	Comments
12.1	More penetrative wtte ✓	1	
12.2	More likely to be absorbed and/or kill bacteria ✓	1	Accept more ionising
12.3	Uses $I_1 r_1^2 = I_2 r_2^2$ ✓ $9.0 \times 10^{-2} \text{ (W m}^{-2}\text{)}$ ✓	2	Accept even with only one value of $I_n r_n^2$ seen
12.4	At 10.6 years, intensity = $1.0 \times 10^{-2} \text{ W m}^{-2}$ ✓ At 15.9 years, intensity $5.0 \times 10^{-3} \text{ W m}^{-2}$ ✓ Ticks box for 10.6 to 15.9 years cao ✓ or uses appropriate equation to calculate time taken for intensity to fall to $8.0 \times 10^{-3} \text{ W m}^{-2}$ 12.3 y or 2.32 half lives (based on $4.0 \times 10^{-2} \text{ W m}^{-2}$) or 18.5 y (based on $9.0 \times 10^{-2} \text{ W m}^{-2}$) Ticks box for 10.6 to 15.9 years cao ✓	3	For the 1 st and 2 nd marking points accept working based on $9.0 \times 10^{-2} \text{ W m}^{-2}$ instead of $4.0 \times 10^{-2} \text{ W m}^{-2}$: At 15.9 years, intensity = $1.1 \times 10^{-2} \text{ W m}^{-2}$ ✓ At 21.2 years, intensity $5.6 \times 10^{-3} \text{ W m}^{-2}$ ✓

12.5	The idea that the high energy electrons would have insufficient penetrative power to reach the bottom of the box (absorbed by box and/or contents) wtte ✓	1	Not just insufficient penetrative power.
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Question	Key
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24	C
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25	A
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26	A
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Question	Answers	Additional comments/Guidelines	Mark	ID details
01	Any two: Do not handle source directly / use tongs to handle source ✓ Store source in lead box or keep source in lead box when not in use or limit exposure time ✓ Do not point source towards people ✓ Display warning notices ✓	Answers must be consistent with sources being used Do not accept any reference to clothing including gloves and goggles. Allow description for 'tongs' Allow misspelling of 'tongs' Do not accept 'keep' without qualification Do not accept keep source away from people/work at a safe distance Accept reference to correct record keeping	2	E
Total			2	

Question	Answers	Additional comments/Guidelines	Mark	ID details
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Question	Answers	Additional comments/Guidelines	Mark	AO
01	Kinetic energy increases (from zero), then decreases ✓ Idea that potential energy decreases, then increases ✓ <u>Elastic</u> potential energy decreases ✓ <u>Gravitational</u> potential energy increases (always) ✓	If no other mark given, allow 1 mark for a correct description of part of the movement (e.g. from X to O, or O to Y). Award MP2 if MP1, MP3 and MP4 given. For MP3 allow elastic PE increases if spring is described as compressing between O and Y Penalise contradictions within answer.	4	AO1 AO1 AO1 AO1
Total			4	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Marking guidance	Mark	Comments
2	11 and 5✓ ν ✓	2	Accept ν or ν_e but not $\bar{\nu}$ Condone (electron) neutrino but not $e\nu$ on its own. Condone νe
Total		2	

Question	Marking guidance	Mark	Comments
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Question	Marking guidance	Mark	Comments
08.1	Charge = $79 \times 1.60 \times 10^{-19}$ (= 1.26×10^{-17} C) OR Mass = $197 \times 1.67 \times 10^{-27}$ (= 3.29×10^{-25} kg) ✓ (Specific charge = charge / mass = $1.26 \times 10^{-17} / 3.29 \times 10^{-25}$) = 3.84×10^7 ✓ C kg⁻¹ ✓	3	Award 1 mark for 79/197 If no marks are awarded, award 1 for specific charge = charge / mass Some working needs to be seen in order to award unit mark.

Question	Marking guidance	Mark	Comments
08.2	The count will be much less at position 2 wtte ✓	1	Less on its own is not sufficient

Question	Marking guidance	Mark	Comments
08.3	The nucleus is (very) small (compared to the distance between nuclei) ✓ The nucleus is (very) dense ✓	2	Accept the nucleus is positively charged Condone the nucleus has the same sign of charge as an alpha particle Allow most of the mass is concentrated in the nucleus but treat "nucleus has a high mass" or "lots of space between nucleus and electrons" as neutral.

Question	Marking guidance	Mark	Comments
08.4	Any pathway that shows a reduced anticlockwise angle of scattering and does not cross the original line, or equivalent description in words ✓	1	

Question	Marking guidance	Mark	Comments
08.5	(Number of protons is less so) less positive charge ✓ Weaker force of (electrostatic) repulsion ✓	2	MP2 cannot be based on incorrect physics.

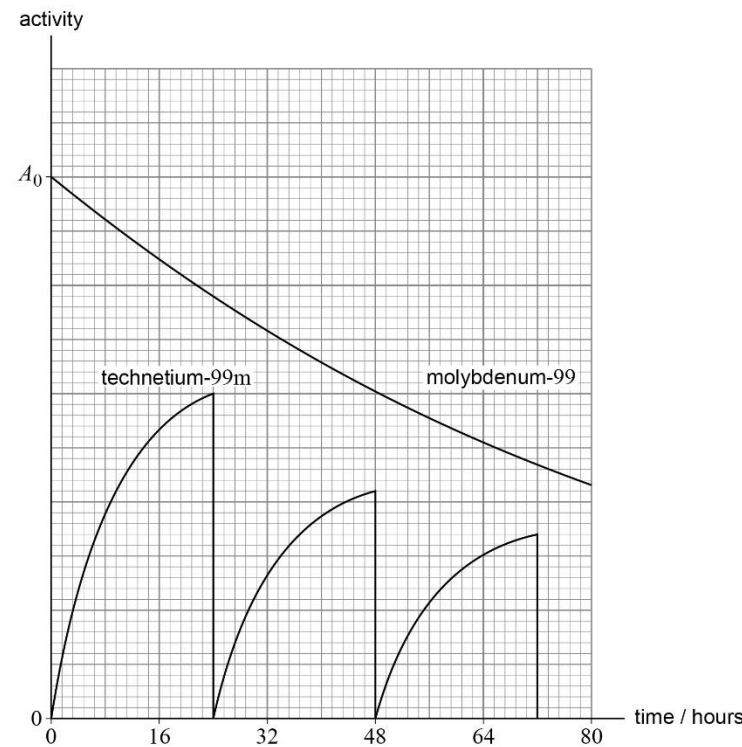
Total		9	
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Question	Marking guidance	Mark	Comments
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Question	Marking guidance	Mark	Comments
09.1	Gamma ✓	1	Allow symbol

Question	Marking guidance	Mark	Comments
09.2	The generator/storage container are shielded with lead ✓ Reduces the amount of radiation that the user is exposed (to a safe level) ✓	2	Accept lead absorbs some of the gamma radiation or wtte for MP2 Accept idea that technetium cannot escape or that it is all contained

Question	Marking guidance	Mark	Comments
09.3	66 hours ✓	1	Accept 65–66 hours

Question	Marking guidance	Mark	Comments
09.4	Peaks at 48 and 72 hours ✓ Extra 2 peaks decreasing in height ✓	2	

Question	Marking guidance	Mark	Comments
09.5	The technetium should be extracted when at/near its maximum activity.✓ This is reached after 24 hours✓	2	Accept reverse argument
Total		8	

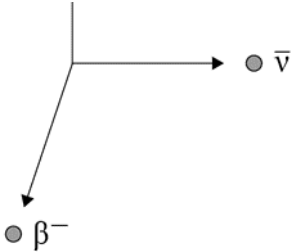
Question	Marking guidance	Mark	Comments
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21	A	can be deflected by a strong magnetic field.
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22	C	60 counts per minute
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23	C	50 counts per minute
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24	A
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Question	Answers	Additional comments/Guidelines	Mark	AO
03.1	Count rate – background count rate ✓	Reject answers that give no idea of rate.	1	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
03.2	Any half value taken and seen on graph ✓ Mean of at least two determinations with answer between 5.6 and 5.7 (h) ✓	For MP1 allow use of activity equation to determine half life.	2	AO3
Total			3	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
04.1	(Experiment must be conducted in vacuum) so that alpha particles reach gold foil OR (In a vacuum) the alpha particles are not stopped by air ✓	Accept reference to range if correct. Accept alternatives for 'stopped'.	1	AO4

Question	Answers	Additional comments/Guidelines	Mark	AO
04.2	Mention of scintillation	Allow description of scintillation. 'Photons produced' is insufficient for idea of flashes of light. Treat references to other detectors/materials as neutral.	1	AO4

Question	Answers	Additional comments/Guidelines	Mark	AO
04.3	Idea that angle (relative to incident direction) and number of particles detected (at that angle) are measured ✓	Accept number of flashes (per second) at different angles	1	AO4
Total			3	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
05.1	Neutron (in the nucleus) decays ✓ ...to form a proton and an electron (and an electron antineutrino) ✓	The first mark point is for the idea that a neutron changes Accept beta particle for electron Accept symbols Equation gets both marks. Condone missing/incorrect neutrino.	2	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
05.2	$A = 90; Z = 39$ cao ✓	Both answers must be correct for the mark to be awarded	1	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
05.3	MAX 2 measurements showed a range of (kinetic) energies/ energies up to a maximum ✓ idea that the energy released in each decay should be the same... ✓ ...the (anti) neutrino was hypothesised to account for the “missing” energy ✓	Allow MP3 if energy of neutrino is linked to the law of conservation of energy.	2	1 × AO1 1 × AO2
Total			5	

Question	Answers	Additional comments/Guidelines	Mark	AO
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22	B	2	AO1
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23	B	4.5 h	AO1
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24	D	gamma	Gamma emissions can escape from the patient and be detected outside.	AO1
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Question	Answers	Additional comments/Guidelines	Mark	AO
05.1	idea that paper does not absorb gamma (sufficiently to change the count rate) ✓ idea that count rate would be the same (regardless of thickness) ✓	MP1 must be in terms of paper ignore any references to safety	2	AO1 AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
05.2	1.67×10^4 to 1.68×10^4 ✓		1	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
05.3	no, because: <i>any two</i> : the activity/count rate decreases (with time) ✓ the old calibration curve would be too high OR if recalibrated, the new curve would be lower ✓ idea that the thickness will be overestimated ✓	do not allow yes it can/could be used for any marks	2	AO2 × 2

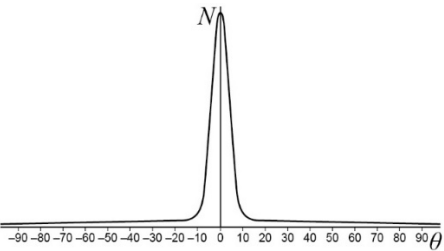
Question	Answers	Additional comments/Guidelines	Mark	AO
05.4	read off for 0.30 mm = 12.2×10^3 (counts s ⁻¹) ✓ divide by 4 = 3.1×10^3 (allow 1 sf) ✓	condone POT error for MP1	2	AO3 × 2

Question	Answers	Additional comments/Guidelines	Mark	AO
05.5	same y -intercept, lower everywhere else, cuts x -axis anywhere before 0.5 mm ✓		1	AO3
Total			8	

Question	Answers	Additional comments/Guidelines	Mark	AO
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24	A	3 min	AO3
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Question	Answers	Additional comments/Guidelines	Mark	AO
08.1	All three points for two marks ✓ ✓ Any two points for one mark ✓ - Central/dense/positively charged nucleus (containing) 13 protons and 14 neutrons 13 electrons and idea that they are in shells/in orbitals/surrounding (the nucleus)		2	AO1 × 2

Question	Answers	Additional comments/Guidelines	Mark	AO
08.2	curve with a single peak at $\theta = 0$ ✓ curve decreases to a small non-zero value at (and beyond) ± 90 degrees ✓	If no scale given, assume centre is zero degrees. Condone $\pm 60^\circ$ as minimum range. Expect to see  Candidate's value must drop by 90% of maximum within an angle of ± 10 degrees	2	AO3 × 2
Total			4	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
09.1	2 from: ✓✓ - subtracts 23 before attempting the inverse-square law - divides any count rate by 9 or multiplies by $\frac{10^2}{30^2}$ - adds 23 to their new count rate 66 ✓	Award one mark for any one answer from the list.	3	AO2 × 3

Question	Answers	Additional comments/Guidelines	Mark	AO
09.2	Suggestion: ✓ Idea that they do not keep hold of the gamma source / place the source in a holder OR hold the source with tongs instead of gloves Reason: ✓ Idea that it is to maximise the distance between user and source / to minimise irradiation	If no other mark given accept for one mark general statement e.g. put a sign on the door. Accept 'fix the source' owtte.	2	AO3 AO4

Question	Answers	Additional comments/Guidelines	Mark	AO
09.3	The mark scheme gives some guidance as to what statements are expected to be seen in a 1- or 2-mark (L1), 3- or 4- mark (L2) and 5- or 6- mark (L3) answer.	For each area, a satisfactory answer is more than half of the statements. A partial answer only includes one statement from that area. Examples of likely statements	6	3 × AO3 3 × AO4
		Other measurements - Take a background count - Subtract background count from count rate (to determine corrected count rate) - Reason: because background (does not vary with distance so) must be eliminated		
		Accuracy - Take repeats of count rate and average - Take background count for a long period of time - Reason: because this reduces (percentage) uncertainty in the corrected count rate		
		Processing - Plot graph of corrected count rate vs $\frac{1}{r^2}$ or equivalent OR log count rate - log distance. - Reason: gradient is a straight line OR gradient is −2 for log-log option OR graph in method would not give a straight line		
Total			11	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
10.1	Momentum argument: water gains/has downward momentum ✓₁ rocket gains/has equal and opposite (upward) momentum with reference to Newton's 3rd Law. ✓₂	Treat references to force on launch tube as neutral Condone ref to Law of conservation of momentum for Newton's 3rd Law in momentum argument. Force alternative: (compressed gas exerts) downwards force on the water ✓₁ an equal and opposite (upward) force is exerted on the (air in the) rocket with ref to Newton's 3rd law ✓₂ Energy alternative: Water leaves bottle and compressed gas expands ✓₁ Compressed gas does work on bottle, bottle gains GPE reference to Law of conservation of energy ✓₂	2	AO2 × 2

Question	Answers	Additional comments/Guidelines	Mark	AO
10.2	At 1.3 s, the gradient stops being positive/is zero ✓ so there is no longer a (resultant) upward force (from the water)✓	For MP1 allow idea that after 1.3 s the velocity starts to decrease OR at 1.3s the velocity is a maximum.	2	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
10.3	Attempt to find area under the graph up to 2.8 s ✓ Area correct in the range of 15–16 squares OR 1 square is equivalent to 2.5 m ✓ Correct answer in the range 37–40 (m) ✓	For MP2 if triangles/trapezium used give mark if at least two areas combined.	3	AO3 × 3

Question	Answers	Additional comments/Guidelines	Mark	AO
10.4	Attempts to find a gradient of a tangent of the steepest part of the graph ✓ Answer in range 32 to 35 (m s^{-2}) ✓	Condone a positive or negative final answer	2	AO3 × 2

Question	Answers	Additional comments/Guidelines	Mark	AO
10.5	Idea that (the resultant force is more than the weight because) both air resistance and weight are acting downwards/in the same direction ✓ Links (greater) acceleration to (greater) force with a reference to $F = ma$ in some form ✓	In MP1 condone idea that the air resistance and weight add together. Alternative: an algebraic form for a eg $a = (mg + R)/m$ ✓ with terms defined ✓	2	AO3
Question	Answers	Additional comments/Guidelines	Mark	AO
10.6	Extend the line at the same terminal velocity ✓ until the area between above the axis is equal to the area below the axis (and then read off the time) owtte ✓	reject the idea that 5.0 s is the time of flight	2	AO3
Total			13	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
03.1	The gamma radiation is detected at a distance inside the detector OR The gamma radiation is emitted at a distance inside the source holder ✓ So the measured values of d will be shorter than the true value ✓	Ignore any random errors MP2 depends on MP1 If no other mark awarded allow 1 mark for the idea that the correct measurement is inside the gamma source to inside the detector	2	AO4

Question	Answers	Additional comments/Guidelines	Mark	AO
03.2	Subtract the background count rate ✓ Repeat and calculate a mean ✓	Allow increase time of measurements	2	AO4
Total			4	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
05.1	$\begin{array}{c} 99 \\ \hline 42 \end{array} \text{Mo} \rightarrow \begin{array}{c} 99\text{m} \\ \hline 43 \end{array} \text{Tc} + \begin{array}{c} 0 \\ \hline -1 \end{array} \beta^- + \begin{array}{c} \bar{\nu}_e \end{array}$ All numbers correct ✓ $\bar{\nu}_{(e)}$ ✓	Do not accept incorrect number on the neutrino for MP1 Do not accept 99m for Mo accept $\bar{\nu}$ but not ν	2	AO1 × 1 AO2 × 1

Question	Answers	Additional comments/Guidelines	Mark	AO
05.2	(Technetium) nucleus is in an excited state ✓ The idea that the resulting gamma emission is not immediate. ✓	Do not allow atom for nucleus Condone isotope or nuclide for nucleus allow excess energy in nucleus	2	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
05.3	Uses consecutive half-lives ✓ Determines number of half-lives to give time taken = 48 (hours) ✓	allow determination of λ and reference to $A = A_0 e^{-\lambda t}$ for MP1	2	AO2 × 1 AO3 × 1

Question	Answers	Additional comments/Guidelines	Mark	AO
05.4	Max 2 from: ✓ ✓ Half-life which is long enough for the patient to be examined Half-life which is short enough that it doesn't remain active in the patient for a long time and damage the body Emits gamma so is detectable Not harmful to patients because (low-energy) gamma emitted (no beta or alpha)	Allow weak beta emitted Accept low toxicity argument	2	AO1
Total			8	

Question	Answers	Additional comments/Guidelines	Mark	AO
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12	C	$4.18 \times 10^7 \text{ C kg}^{-1}$	AO3
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18	B	α, β^-, β^-	AO1
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22	C	Very few of the alpha particles were deflected through a large angle.	AO1
			AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
04.1	Determines A ($= 5$) and Z ($= 2$) for X ✓ Correct use of $\frac{q}{m}$ to determine specific charge of X or helium nucleus ✓ 9.6×10^6 (C kg⁻¹) ✓		3	1 × AO1 1 × AO2 1 × AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
04.2	Any two from ✓✓ • Move in opposite directions... • ... with the neutron having the same magnitude of momentum as the helium-4 nucleus OR (So that) momentum is conserved • Decay energy transferred to the kinetic energies of the particles Speed of neutron is 4 times greater than the speed of the helium-4 nucleus (ignoring relativistic effects) ✓		3	1 × AO1 2 × AO2
Total			6	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO														
06.1	The mark scheme gives some guidance as to what statements are expected in a 1- or 2-mark (L1), 3- or 4-mark (L2) or 5-mark (L3) answer. Guidance provided on page 3 of the Mark Scheme should be used to assist in marking this question. <table><tr><th>Mark</th><th>Criteria</th></tr><tr><td>5</td><td>A fair attempt to analyse all three areas in detail. 5 marks can be awarded even if there is an error and/or parts of an aspect missing.</td></tr><tr><td>4</td><td>A attempt to analyse all three areas with at least one in detail.</td></tr><tr><td>3</td><td>Attempt made to discuss more than one area, and one area covered in detail or an attempt to discuss all three areas. Whilst there will be gaps, there should be only an occasional error.</td></tr><tr><td>2</td><td>Attempt made to discuss more than one area, or one area covered in detail. There are likely to be several errors or omissions in the discussion.</td></tr><tr><td>1</td><td>Any relevant physics (partial coverage of one area).</td></tr><tr><td>0</td><td>No relevant analysis.</td></tr></table>	Mark	Criteria	5	A fair attempt to analyse all three areas in detail. 5 marks can be awarded even if there is an error and/or parts of an aspect missing.	4	A attempt to analyse all three areas with at least one in detail.	3	Attempt made to discuss more than one area, and one area covered in detail or an attempt to discuss all three areas. Whilst there will be gaps, there should be only an occasional error.	2	Attempt made to discuss more than one area, or one area covered in detail. There are likely to be several errors or omissions in the discussion.	1	Any relevant physics (partial coverage of one area).	0	No relevant analysis.	The following statements are likely to be present: Making the necessary measurements • Use of a suitable detector (type of detector need not be named) • Way of measuring count rate eg ratemeter or scalar counter and stop-clock • Means of measuring distance between source and detector • Measures the count rate at a range of distances with at least 5 distances used • Measure and deduct the background count rate Performing the experiment safely and accurately • Eliminates alpha radiation using a suitable named filter or minimum distance greater than 5.0 cm. • Repeats and averages or counts for a suitably long period (to reduce uncertainties) • Suitable range of distances with max at least $4 \times \text{min}$ (max may be $< 1 \text{ m}$ due to limited intensity of source) • One suitable safety precaution such as appropriate handling technique, keeping source in suitable container when not in use, avoid directing source towards people etc Processing results to demonstrate the inverse-square law graphically • Plots C against $\frac{1}{d^2}$ (accept $\sqrt{C}$ against $\frac{1}{d}$ or $\log C$ against $\log d$) • Inverse-square law demonstrated if the graph is a straight line through the origin (accept and reward comments about intercept relating to position of detection within the detector)	5	AO4
Mark	Criteria																	
5	A fair attempt to analyse all three areas in detail. 5 marks can be awarded even if there is an error and/or parts of an aspect missing.																	
4	A attempt to analyse all three areas with at least one in detail.																	
3	Attempt made to discuss more than one area, and one area covered in detail or an attempt to discuss all three areas. Whilst there will be gaps, there should be only an occasional error.																	
2	Attempt made to discuss more than one area, or one area covered in detail. There are likely to be several errors or omissions in the discussion.																	
1	Any relevant physics (partial coverage of one area).																	
0	No relevant analysis.																	

Question	Answers	Additional comments/Guidelines	Mark	AO
06.2	Any one from ✓ - Makes one half-life determination correctly from graph - Multiplies half-life by 4 Makes at least two determinations and averages ✓ 5.4×10^{10} (s) ✓	Allow POT error in MP1 and MP2 Expect to see $1.3 \times 10^{10} - 1.4 \times 10^{10}$ Allow mp2 for candidates who determine the time taken for the activity to fall to $\frac{1}{4}$ or $\frac{1}{8}$ of original. Accept answer in the range 5.2×10^{10} to 5.6×10^{10} (s)	3	$1 \times \text{AO1}$ $2 \times \text{AO3}$
Total			8	

Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
09.1	${}^3_1\text{H} \rightarrow {}^3_2\text{Z} + {}^0_{-1}\beta^- + {}^0_0\bar{\nu}_e$ Helium and electron numbers correct ✓		1	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
09.2	Any two from ✓✓ • Correct read-off in the range 2.5 to 2.7 keV • Converts their value to joules • Use of $E_k = \frac{1}{2}mv^2$ with their converted value Final answer in the range that rounds to $3.0 \times 10^7 \text{ m s}^{-1}$ or $3.1 \times 10^7 \text{ m s}^{-1}$ ✓	Allow POT error in MP1 and MP2 expect to see $4.16 \times 10^{-16} \text{ J}$	3	1 × AO1 1 × AO2 1 × AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
09.3	(There is a range of beta particle kinetic energies) Idea that the sum of the beta particle energy and the energy of the neutrino is constant ✓ Energy (and momentum) must be conserved ✓ Idea that if there is a range of kinetic energies then another particle must be carrying away some energy ✓	Allow the same amount of energy is released in each decay	3	1 × AO1 2 × AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
09.4	Minimum energy (close to) zero and Maximum energy 18.6 (keV) ✓	condone zero energy allow 18.4 – 18.8 (keV)	1	AO3
Total			8	

Question	Answers	Additional comments/Guidelines	Mark	AO
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17	B	2.1×10^8	AO2
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24	D	600	300	225	75	AO3
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25	A	4	AO3
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Total: 14 marks

Question	Answers	Additional comments/Guidelines	Mark	AO
04.1	proton, electron and antineutrino ✓		1	AO1

Question	Answers	Additional comments/Guidelines	Mark	AO
04.2	annihilation ✓ releasing two gamma photons only ✓		2	AO1

Total			3	
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Question	Answers	Additional comments/Guidelines	Mark	AO
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Question	Answers	Additional comments/Guidelines	Mark	AO
07.1	Uses $2 \times 1.6 \times 10^{-19}$ AND divides by either $2m_p + 2m_n$ OR $4 \times 1.67 \times 10^{-27}$ ✓ Uses $\frac{\text{charge}}{\text{mass}}$ ✓ 4.79×10^7 (C kg ⁻¹) ✓	Allow use of $4 \times 1.66(1) \times 10^{-27}$ leading to 4.82×10^7 (C kg ⁻¹)	3	2 × AO1 1 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
07.2	Idea that the activity will not drop to the level where it needs to be replaced OR Will not alter count rate during demonstration ✓	Condone activity will be low so risk will be low	1	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
07.3	Idea that collisions with air particles reduces velocity / momentum / kinetic energy ✓		1	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
07.4	Any two from: ✓✓ • measure the count rate with no source present • for a length of time • subtract from the reading when the source is present		2	AO1

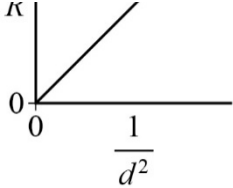
Question	Answers	Additional comments/Guidelines	Mark	AO
07.5	Idea that some of the decay products of radium-226 are radioactive and emit beta and/or gamma ✓ beta and/or gamma with a range greater than alpha ✓		2	1 × AO2 1 × AO3

Total			9	
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Question	Answers	Additional comments/Guidelines	Mark	AO
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11	C	17 000 years	AO3
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12	A	9.8×10^4	AO2
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21	D		AO3
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Question	Answers	Additional comments/Guidelines	Mark	AO
08.1	Description or diagram showing radiation source and detector with material in between ✓₁ At least one of the following points: ✓₂ All three of the following points: ✓₂ ✓₃ • if radiation is blocked by paper or several cm of air then alpha • if radiation is not blocked by several mm of aluminium then gamma • if radiation is not blocked by paper or several cm of air but is blocked by aluminium then beta	Do not allow an uncorrected count rate to decrease to 0 (OWTTE) for ✓₃ but condone for ✓₂ Alternative using PH03 physics: Allow use of a magnetic or electric field with a movable detector ✓₁ At least one of the following points: ✓₂ First three or first and last of the following points: ✓₂ ✓₃ • if undeflected gamma • with E field, alpha is detected towards negative plate • with E field beta is detected towards positive plate • with B field deflection consistent with FLHR	3	AO4

Question	Answers	Additional comments/Guidelines	Mark	AO
08.2	Any two from: ✓ • use long tongs to handle source • do not point the source towards people • keep sources in a lead-lined box when not in use • display warning signs • keep suitable distance (e.g. 2 m) away from source when not moving source Accept reference to correct record-keeping	Allow other sensible precautions Do not allow use goggles / radiation suit	1	AO4

Question	Answers	Additional comments/Guidelines	Mark	AO
08.3	The barium-137m nucleus is in an excited state ✓ Idea that gamma emission / de-excitation is not immediate ✓ (and) after 153 seconds half of the excited nuclei will have de-excited / decayed / released a gamma photon / no longer be excited / no longer be in a metastable state ✓	Condone missing nucleus but do not allow atom or isotope Condone gamma ray or gamma for gamma photon Allow answers that refer to number remaining Allow Ba or barium-137 for barium-137m	3	2 × AO1 1 × AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
08.4	Appropriate working leading to an answer of 28 years ✓	Working could be seen on the graph	1	AO2

Total			8	
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Question	Answers	Additional comments/Guidelines	Mark	AO
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12	B	electron	AO1
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13	A	X and Y are isotopes.	AO3
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14	D	$4.7 \times 10^3 \text{ kg}$	AO1
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16	C	Each emitted beta particle shares the released energy with another particle.	AO1
			AO1

24	A	8.7 W m^{-2}	AO2
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Unknown Unknown - Unit 1 - Q03

Question	Marking guidance	Mark	Comments
03.1	(electron) <u>anti</u> neutrino ✓	1	
03.2	They annihilate ✓ Emitting two gamma photons ✓	2	

Question	Marking guidance	Mark	Comments
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